

Industrial Internship Report on "Secure Password Manager"

Prepared by

Shah Harsh Umeshbhai

Executive Summary

This report provides details of the Industrial Internship provided by upSkill Campus and The IoT Academy in collaboration with Industrial Partner UniConverge Technologies Pvt. Ltd. (UCT). This internship was focused on developing foundational and advanced Python programming skills over a 4 to 6-week timeframe. The capstone of this internship was the development of a practical, industry-relevant software application.

My project was the **Secure Password Manager**, a desktop application designed to securely encrypt, store, and manage user credentials locally. This internship gave me a highly valuable opportunity to get exposure to industrial problem-solving, UI/UX design considerations, and the implementation of real-world cryptographic security measures. It was an overall great experience that effectively bridged the gap between academic theory and practical software development.

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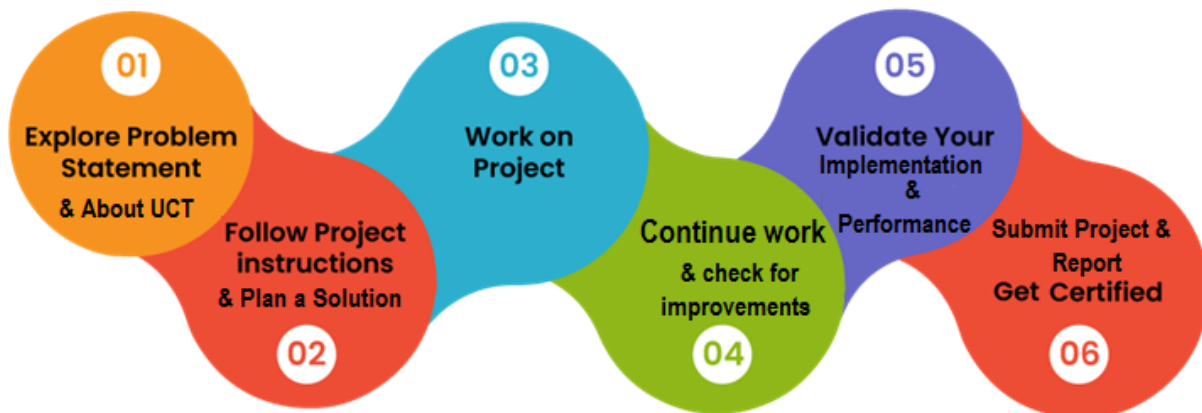
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1 Preface

Summary of the Work: Over the course of the internship, the learning progression moved from mastering fundamental Python data structures and control flow to utilizing advanced external libraries. The culmination of these weeks was the architecture and deployment of a fully functional graphical application.

Need for Relevant Internship: In the rapidly evolving tech landscape, theoretical knowledge is insufficient without practical application. This internship was crucial for career development as it provided hands-on experience with standard industry practices, debugging, and project architecture.

Brief about the Project: The problem statement involved addressing the insecurity of storing plain-text passwords. The developed solution, the Secure Password Manager, utilizes Python's tkinter library for a seamless user interface and the cryptography library for robust data encryption, ensuring local credentials cannot be compromised. Opportunity given by USC/UCT.



Opportunity & Planning: I am highly grateful for the opportunity provided by upSkill Campus and UniConverge Technologies. The program was strategically planned, initiating with exploratory problem statements and learning modules, followed by structured project instructions, validation of the implemented code, and final report submission.

2 Introduction

2.1 About UniConverge Technologies Pvt Ltd

A company established in 2013 and working in Digital Transformation domain and providing Industrial solutions with prime focus on sustainability and RoI.

For developing its products and solutions it is leveraging various **Cutting Edge Technologies** e.g. **Internet of Things (IoT), Cyber Security, Cloud computing (AWS, Azure), Machine Learning, Communication Technologies (4G/5G/LoRaWAN), Java Full Stack, Python, Front end** etc.



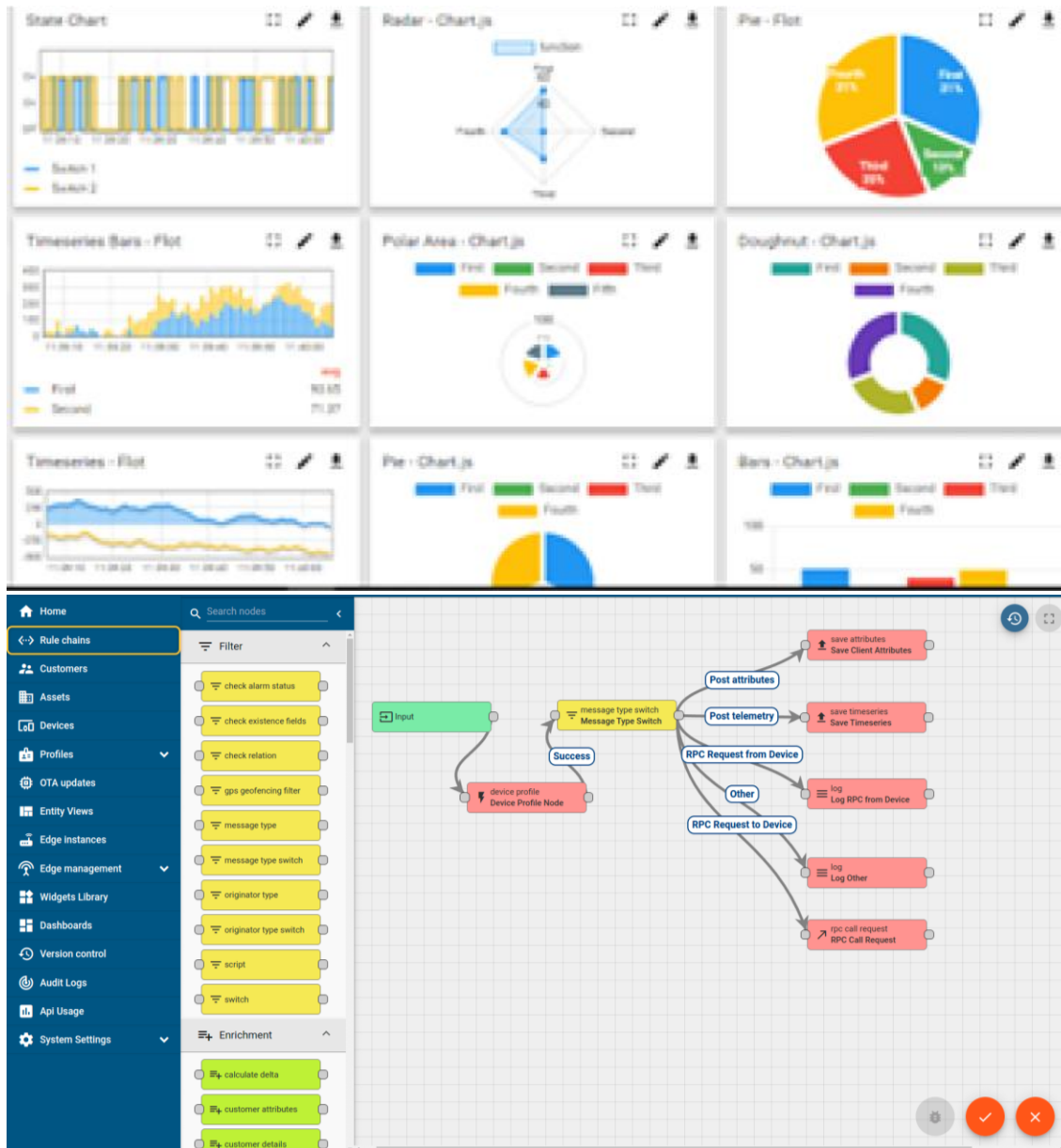
i. UCT IoT Platform ()

UCT Insight is an IOT platform designed for quick deployment of IOT applications on the same time providing valuable “insight” for your process/business. It has been built in Java for backend and ReactJS for Front end. It has support for MySQL and various NoSql Databases.

- It enables device connectivity via industry standard IoT protocols - MQTT, CoAP, HTTP, Modbus TCP, OPC UA
- It supports both cloud and on-premises deployments.

It has features to

- Build Your own dashboard
- Analytics and Reporting
- Alert and Notification
- Integration with third party application(Power BI, SAP, ERP)
- Rule Engine



FACTORY WATCH

ii. Smart Factory Platform ()

Factory watch is a platform for smart factory needs.

It provides Users/ Factory

- with a scalable solution for their Production and asset monitoring
- OEE and predictive maintenance solution scaling up to digital twin for your assets.
- to unleash the true potential of the data that their machines are generating and helps to identify the KPIs and also improve them.
- A modular architecture that allows users to choose the service that they want to start and then can scale to more complex solutions as per their demands.

Its unique SaaS model helps users to save time, cost and money.



Machine	Operator	Work Order ID	Job ID	Job Performance	Job Progress		Output		Rejection	Time (mins)				Job Status	End Customer
					Start Time	End Time	Planned	Actual		Setup	Pred	Downtime	Idle		
CNC_S7_81	Operator 1	WO0405200001	4168	58%	10:30 AM		55	41	0	80	215	0	45	In Progress	i
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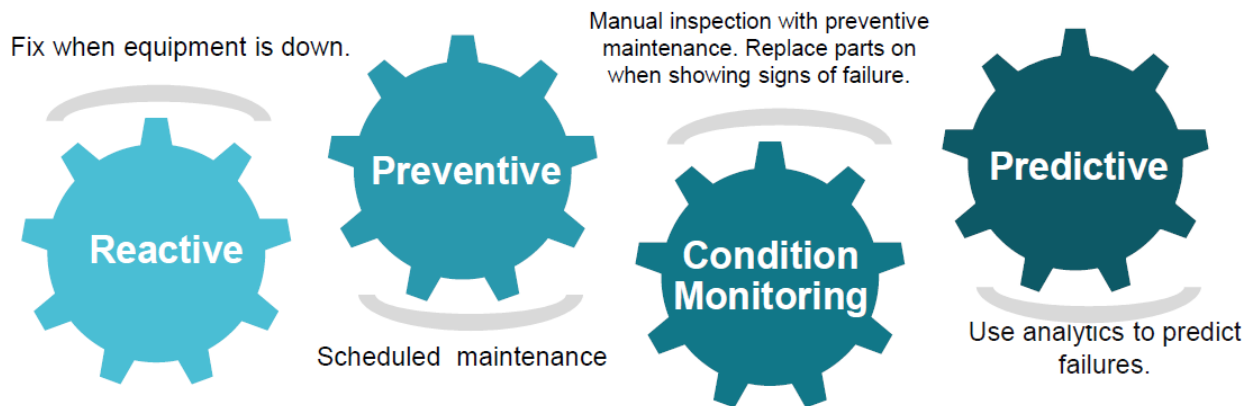


iii. LoRaWAN based Solution

UCT is one of the early adopters of LoRAWAN teschnology and providing solution in Agritech, Smart cities, Industrial Monitoring, Smart Street Light, Smart Water/ Gas/ Electricity metering solutions etc.

iv. Predictive Maintenance

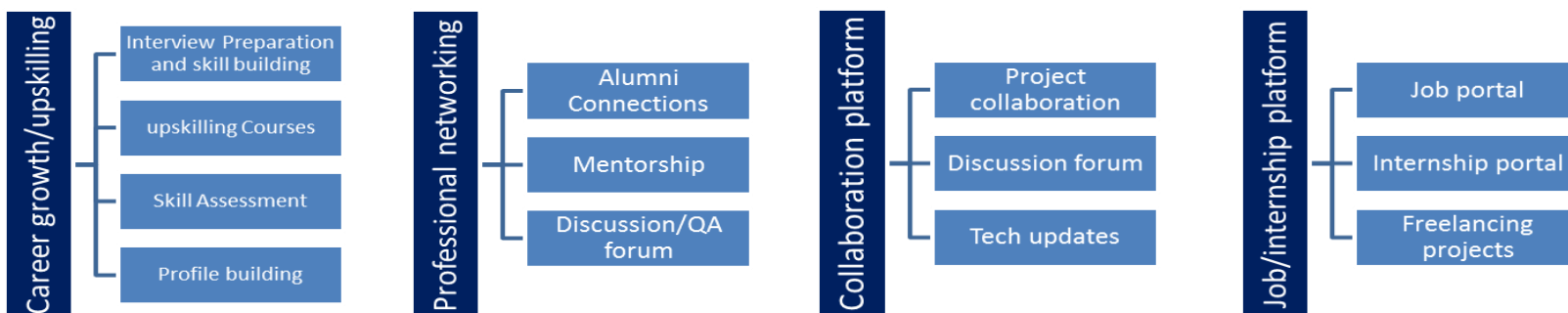
UCT is providing Industrial Machine health monitoring and Predictive maintenance solution leveraging Embedded system, Industrial IoT and Machine Learning Technologies by finding Remaining useful life time of various Machines used in production process.



2.2 About upskill Campus (USC)

upskill Campus along with The IoT Academy and in association with Uniconverge technologies has facilitated the smooth execution of the complete internship process.

USC is a career development platform that delivers **personalized executive coaching** in a more affordable, scalable and measurable way.



2.3 The IoT Academy

The IoT academy is EdTech Division of UCT that is running long executive certification programs in collaboration with EICT Academy, IITK, IITR and IITG in multiple domains.

2.4 Objectives of this Internship program

The objective for this internship program was to

- get practical experience of working in the industry.
- to solve real world problems.
- to have improved job prospects.
- to have Improved understanding of our field and its applications.
- to have Personal growth like better communication and problem solving.

2.5 Reference

- [1] Python Software Foundation. Python Official Documentation.
- [2] Python Cryptographic Authority (PyCA). "Cryptography" Library Documentation.
- [3] Tkinter — Python interface to Tcl/Tk Documentation.

2.6 Glossary

Terms	Acronym
GUI	Graphical User Interface (The visual window users interact with).
Fernet	A symmetric encryption standard used to scramble data securely.
JSON	JavaScript Object Notation (A lightweight format for storing structured data).
Vault	The secure, encrypted local database file where credentials are saved.

3 Problem Statement

In the modern digital landscape, individuals are required to manage dozens of complex passwords across various accounts. Storing these passwords in plain text, writing them down, or reusing weak passwords poses a significant cybersecurity risk. There is a critical need for an offline, self-contained solution that empowers users to manage their credentials securely without relying on third-party cloud services.

The assigned problem statement was to architect and develop a **Secure Password Manager** application. The scope of this project required designing a Graphical User Interface (GUI) to collect and display data, implementing a random generator to create highly secure passwords, and utilizing standard encryption algorithms to ensure that the saved data remains completely unreadable to unauthorized users.

4 Existing and Proposed solution

Existing Solutions and Their Limitations: Currently, users either rely on insecure methods—such as reusing identical passwords or storing them in unencrypted text files—or they use commercial cloud-based password managers. The limitation of cloud-based solutions is that they represent centralized targets for massive cyberattacks. Furthermore, they often require active internet connections to sync and frequently charge premium subscription fees for advanced features.

Proposed Solution: The proposed solution is a localized, desktop-based **Secure Password Manager**. Developed in Python, it utilizes a tkinter graphical interface to collect user credentials and the cryptography library (specifically the Fernet module) to encrypt the data before saving it offline in a local JSON database.

Value Addition: This system provides the user with complete sovereignty over their data. By keeping the encrypted database strictly on the local machine, it eliminates the risk of remote server breaches. It also adds significant value through its built-in random password generator, ensuring users can instantly create highly secure credentials without relying on external websites.

4.1 Code submission (Github link)

<https://github.com/trial0743210/upskillcampus/blob/main/Password%20manager/PasswordManager.py>
.py

4.2 Report submission (Github link) :

https://github.com/trial0743210/upskillcampus/blob/main/PasswordManager_ShahHarsh_USC_UCT.pdf

5 Proposed Design/ Model

Given more details about design flow of your solution. This is applicable for all domains. DS/ML Students can cover it after they have their algorithm implementation. There is always a start, intermediate stages and then final outcome.

5.1 High Level Diagram (if applicable)

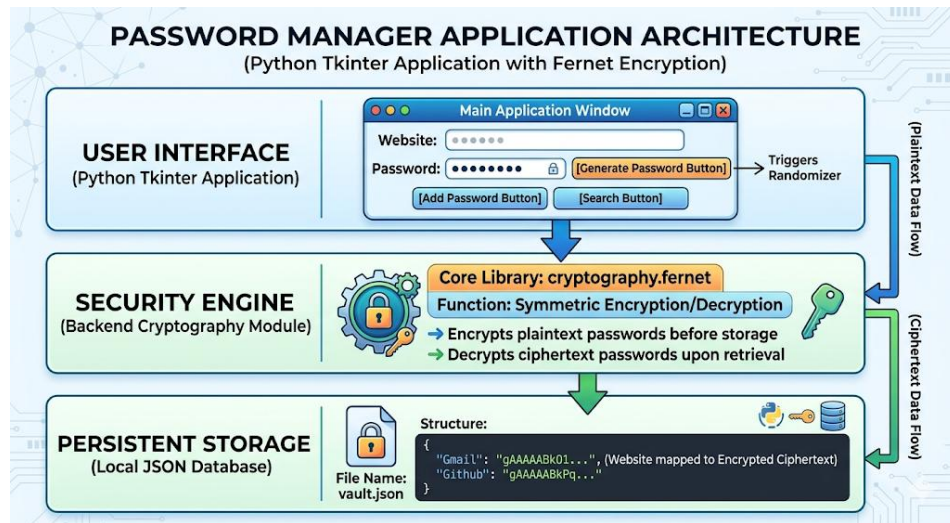


Figure 1: HIGH LEVEL DIAGRAM OF THE SYSTEM

6 Performance Test

Performance testing is a critical phase to ensure the application is robust enough for real-world use, moving it beyond a mere academic script.

Design Constraints & Handling: The primary constraints for this desktop application were operational stability (preventing crashes from bad user input) and cryptographic speed (MIPS/processing time).

- **Input Constraint:** Users might leave fields blank or enter unsupported characters. This was handled using strict try-except blocks and input validation within the tkinter logic.
- **Speed/Memory Constraint:** Local file I/O operations and Fernet encryption needed to be instantaneous to provide a good user experience. Python's built-in json library handles small dictionary sizes with negligible memory overhead.

6.1 Test Plan/ Test Cases

- **Test Case 1 (Authentication):** Enter an incorrect Master Password upon launch. (Expected: Access denied).
- **Test Case 2 (Input Validation):** Attempt to save a password without entering a Website name. (Expected: Error popup, no crash).
- **Test Case 3 (Encryption):** Save a valid credential and inspect the backend database. (Expected: Database shows randomized ciphertext, not plain text).

6.2 Test Procedure

The application was executed in a standard Windows Python 3.x environment. The Master Password gate was tested first. Following successful entry, the random generator was triggered to create a 12-character string. This string, along with a test website name, was submitted via the GUI. Finally, the `vault.json` file was opened in a standard text editor to manually verify the cryptographic output.

6.3 Performance Outcome

The application passed all test cases successfully. The try-except loops prevented any system crashes during forced input errors. Furthermore, the `cryptography.fernet` module processed the encryption and decryption in mere milliseconds, resulting in zero noticeable lag in the GUI, confirming that the system is highly durable and efficient for daily credential management.

7 My learnings

This 4-week internship transformed my theoretical understanding of programming into practical, functional engineering skills.

- **Core Python:** I solidified my understanding of complex data structures, specifically utilizing nested dictionaries to map and manage user data efficiently.
- **UI/UX Development:** By building the project with tkinter, I learned how to architect event-driven applications, wire buttons to backend functions, and manage a graphical main loop.
- **Cybersecurity Principles:** I gained hands-on experience implementing symmetric encryption, learning how to generate, store, and utilize cryptographic keys securely.

This holistic experience—from conceptualizing a solution to debugging a final product—has greatly improved my problem-solving capabilities and prepared me for professional roles in backend development and software engineering.

8 Future work scope

While the current Secure Password Manager is fully functional, there are several features that could be implemented in future iterations to enhance its commercial viability:

1. **Biometric Authentication:** Integrating the application with OS-level security features (like Windows Hello or Touch ID) to replace or supplement the Master Password.
2. **Auto-Fill Browser Extension:** Developing a companion browser extension that communicates securely with the local desktop app to auto-fill credentials on websites.
3. **Opt-in Encrypted Syncing:** Adding an option for users to securely sync their encrypted vault.json file to a personal cloud drive (like Google Drive) to allow access across multiple personal devices without exposing the raw data.